Electromagnetism from Zero

Lecture Notes for Computer Science Students Who Need a Very Slow Start

Prepared as mathematical-physical lecture notes

What this course is about

Electric charge creates electric fields. Moving charge creates magnetic fields. Changing fields create other fields. These ideas explain circuits, sensors, motors, antennas, wireless communication, electromagnetic waves, and many numerical simulations.

The goal is to understand the small set of ideas from which most formulas follow.

Key idea

This text assumes almost no physics background and very little mathematical fluency. It repeats basic vector ideas, explains units, and translates formulas into algorithmic thinking. It is still a physics lecture note: the final goal is to understand real electromagnetism, not only plug numbers into formulas.

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1 How to read these notes

Electromagnetism can look difficult because it combines several things at once:

- vectors, because fields have directions;
- functions, because fields depend on position and time;
- rates of change, because changing fields matter;
- geometry, because angles, areas, and directions matter;
- physical units, because every formula must be dimensionally meaningful.

For many students the hard part is not the physics itself. The hard part is that a formula contains too many symbols at once. Therefore we will use a repeated pattern:

1. Say what the physical situation is.
2. Draw a simple picture.
3. Name the quantities.
4. Write the formula.
5. Substitute numbers carefully.
6. Interpret the answer in words.

Computer science viewpoint

In computer science language, a physical law is often like an API contract. You must know:

- what inputs it expects,
- what output it returns,
- what units are required,
- what assumptions are hidden inside it.

A formula used outside its assumptions is like a function called with invalid arguments.

1.1 The minimum mathematical toolkit

We will use the following mathematical objects.

Scalar

A scalar is a single number with a unit. Examples: mass $m = 2\,\mathrm{kg}$, charge $q = 3\,\mathrm{C}$, voltage $V = 5\,\mathrm{V}$, time $t = 10\,\mathrm{s}$.

Vector

A vector has a size and a direction. Examples: force, velocity, electric field, magnetic field.

A two-dimensional vector can be written as

$$\vec{a} = (a_x, a_y) = a_x\hat{\imath} + a_y\hat{\jmath}.$$

Its magnitude is

$$|\vec{a}| = \sqrt{a_x^2 + a_y^2}.$$

For example, if

$$\vec{a} = (3, 4),$$

then

$$|\vec{a}| = \sqrt{3^2 + 4^2} = 5.$$

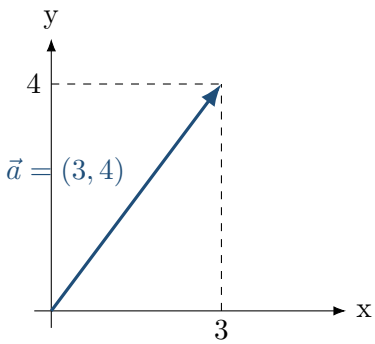


Figure 1: A vector is not just a number. It has components and direction.

1.2 Units are part of the meaning

A formula without units is dangerous. Units tell you what kind of number you have.

Quantity	Symbol	SI unit
Charge	q	coulomb, C
Current	I	ampere, A
Voltage	V	volt, V
Electric field	E	newton per coulomb, N/C, or volt per meter, V/m
Magnetic field	B	tesla, T
Resistance	R	ohm, Ω
Capacitance	C	farad, F
Inductance	L	henry, H

Common mistake

Do not write a bare number when the quantity has a unit. Saying "the voltage is 5" is incomplete. Say 5 V. Saying "the distance is 5" is incomplete. Say 5 m, 5 cm, or another unit.

2 Electric charge: the source of electric effects

2.1 What is charge?

Electric charge is a basic property of matter. There are two signs of charge:

- positive charge,
- negative charge.

The basic rule is:

Key idea

Charges with the same sign repel. Charges with opposite signs attract.

An electron has negative charge. A proton has positive charge. Ordinary matter usually contains almost equal amounts of positive and negative charge, so it looks electrically neutral at large scales.

The unit of charge is the coulomb, written C. The elementary charge is

$$e \approx 1.60 \cdot 10^{-19}\,\mathrm{C}.$$

An electron has charge $-e$, and a proton has charge $+e$.

2.2 Coulomb's law

Two point charges exert forces on each other. If the charges are q_1 and q_2, separated by distance r, the magnitude of the force is

$$F = k\frac{|q_1 q_2|}{r^2},$$

where

$$k = \frac{1}{4\pi\varepsilon_0} \approx 8.99 \cdot 10^9\,\mathrm{N\,m^2/C^2}.$$

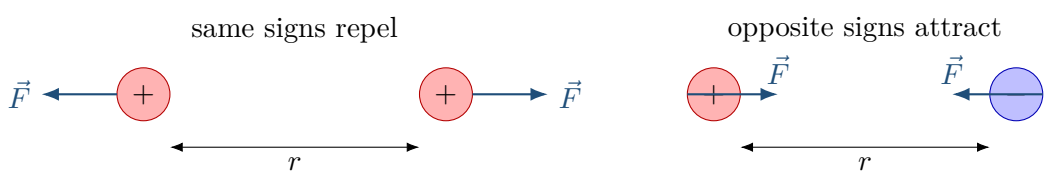


Figure 2: Coulomb force depends on sign and distance.

2.3 Step-by-step example

Example 1: force between two charges

Two charges are

$$q_1 = 2 \cdot 10^{-6}\,\mathrm{C}, \qquad q_2 = 3 \cdot 10^{-6}\,\mathrm{C}.$$

They are separated by

$$r = 0.50\,\mathrm{m}.$$

Find the magnitude of the electric force.

Step 1. Write Coulomb's law:

$$F = k\frac{|q_1 q_2|}{r^2}.$$

Step 2. Substitute the numbers:

$$F = (8.99 \cdot 10^9)\frac{(2 \cdot 10^{-6})(3 \cdot 10^{-6})}{(0.50)^2}.$$

Step 3. Multiply the charges:

$$(2 \cdot 10^{-6})(3 \cdot 10^{-6}) = 6 \cdot 10^{-12}.$$

Step 4. Square the distance:

$$(0.50)^2 = 0.25.$$

Step 5. Compute:

$$F = (8.99 \cdot 10^9)\frac{6 \cdot 10^{-12}}{0.25} \approx 0.216\,\mathrm{N}.$$

The force magnitude is about 0.22 N. Since both charges are positive, the force is repulsive.

Common mistake

The distance is squared. If the distance doubles, the force becomes four times smaller. This is called an inverse-square law.

2.4 Superposition: many charges

If several charges act on one charge, the total force is the vector sum:

$$\vec{F}_{\text{total}} = \vec{F}_1 + \vec{F}_2 + \cdots + \vec{F}_n.$$

This is called the principle of superposition.

Computer science viewpoint

Superposition is like accumulating contributions in a loop:

```
F_total = vector(0, 0, 0)
for source_charge in charges:
    F_total += force_from(source_charge, test_charge)
```

This idea appears again for electric fields, potentials, and numerical simulations.

3 Electric field: force per unit charge

3.1 Why introduce fields?

Coulomb's law describes forces between charges. But modern physics uses a deeper idea: a charge creates an electric field in the space around it. Another charge placed in that field feels a force.

Electric field

The electric field at a point is the force per unit positive test charge:

$$\vec{E} = \frac{\vec{F}}{q}.$$

Therefore,

$$\vec{F} = q\vec{E}.$$

The electric field has units

$$\frac{\mathrm{N}}{\mathrm{C}}.$$

Key idea

A field is a value assigned to every point in space. A temperature field assigns a temperature to each point. An electric field assigns a vector to each point.

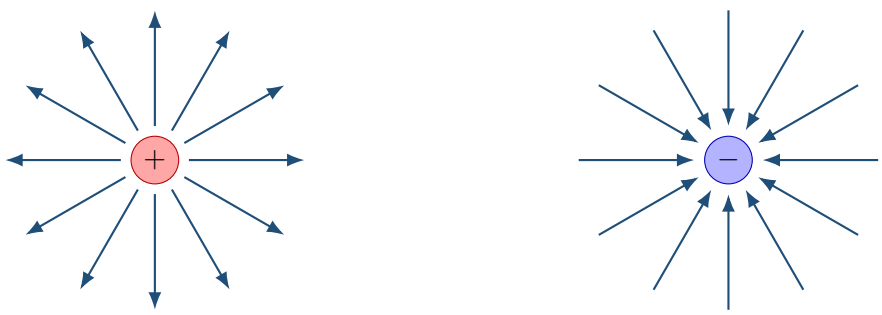
3.2 Electric field of a point charge

A point charge Q creates an electric field whose magnitude at distance r is

$$E = k\frac{|Q|}{r^2}.$$

The direction is:

- away from a positive charge,
- toward a negative charge.



positive charge: field points outward negative charge: field points inward

Figure 3: Electric field direction around point charges.

3.3 Field as a function

In one dimension, a field can be written as $E(x)$. In three dimensions, it is a vector-valued function:

$$\vec{E}(x, y, z) = \big(E_x(x, y, z), E_y(x, y, z), E_z(x, y, z)\big).$$

Computer science viewpoint

For a programmer, a field can be represented as:

```
E[x][y][z] = Vector3(Ex, Ey, Ez)
```

In a simulation, space is often replaced by a grid. At each grid point we store a vector.

3.4 Example: field and force

Example 2: force from a known electric field

An electric field is

$$\vec{E} = (200, 0)\,\mathrm{N/C}.$$

A charge

$$q = 3 \cdot 10^{-6}\,\mathrm{C}$$

is placed in this field. Find the force.
Using $\vec{F} = q\vec{E}$,

$$\vec{F} = (3 \cdot 10^{-6})(200, 0) = (6 \cdot 10^{-4}, 0)\,\mathrm{N}.$$

So the force points in the positive x-direction and has magnitude

$$6 \cdot 10^{-4}\,\mathrm{N}.$$

If the charge were negative, the force would point in the opposite direction.

3.5 Uniform electric field

A uniform electric field has the same magnitude and direction everywhere. Between two large parallel plates, the field is approximately uniform.

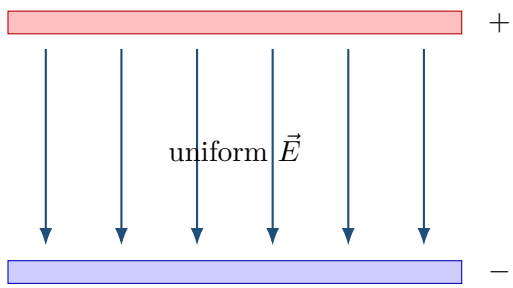


Figure 4: Approximate uniform electric field between parallel plates.

4 Electric potential and voltage

4.1 Why voltage is easier than field

Electric field is a vector. It has direction. Voltage is a scalar. It is only a number at each point. This often makes voltage easier to use, especially in circuits.

Electric potential

Electric potential V is electric potential energy per unit charge:

$$V = \frac{U}{q}.$$

The unit is the volt:

$$1\,\mathrm{V} = 1\frac{\mathrm{J}}{\mathrm{C}}.$$

Voltage difference is what matters in most applications:

$$\Delta V = V_B - V_A.$$

4.2 Energy and voltage

If a charge q moves through a voltage difference ΔV, the change in electric potential energy is

$$\Delta U = q\Delta V.$$

Example 3: energy from voltage

A charge $q = 2\,\mathrm{C}$ moves through a voltage difference of $5\,\mathrm{V}$. The energy change is

$$\Delta U = q\Delta V = (2)(5) = 10\,\mathrm{J}.$$

This means $10\,\mathrm{J}$ of energy is associated with this movement.

4.3 Relation between field and voltage

In one dimension, the electric field is related to voltage by

$$E_x = -\frac{\mathrm{d}V}{\mathrm{d}x}.$$

This means:

Key idea

The electric field points in the direction where voltage decreases fastest.

If the voltage changes linearly between two plates separated by distance d, then approximately

$$E \approx \frac{|\Delta V|}{d}.$$

Example 4: field between plates

Two plates have voltage difference 12 V and are separated by 0.02 m. Estimate the electric field magnitude.

$$E \approx \frac{12}{0.02} = 600\,\mathrm{V/m}.$$

Since $1\,\mathrm{V/m} = 1\,\mathrm{N/C}$, this is also $600\,\mathrm{N/C}$.

4.4 Equipotential lines

An equipotential line is a set of points with the same voltage. Moving along an equipotential line does not change electric potential energy.

Electric field lines are perpendicular to equipotential lines.

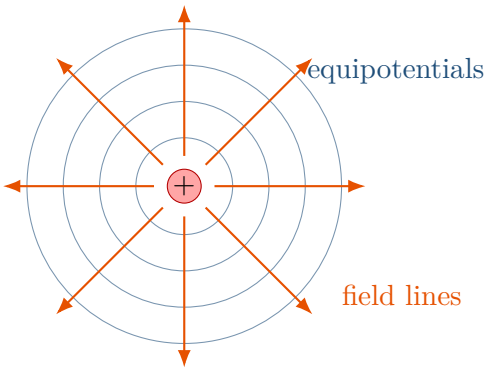


Figure 5: For a positive point charge, equipotentials are circles/spheres and field lines are radial.

5 Capacitors: storing electric energy

5.1 What is a capacitor?

A capacitor is a device that stores separated electric charge. The simplest ideal capacitor consists of two conducting plates separated by an insulator.

Capacitance

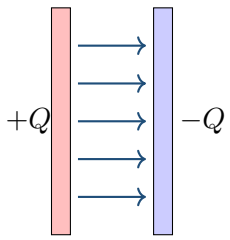
Capacitance measures how much charge is stored per voltage:

$$C = \frac{Q}{V}.$$

Equivalently,

$$Q = CV.$$

The unit is the farad, F.



electric field between plates

Figure 6: A capacitor stores separated charge and electric field energy.

5.2 Energy stored in a capacitor

The energy stored in a capacitor is

$$U = \frac{1}{2}CV^2.$$

Equivalent forms are

$$U = \frac{Q^2}{2C}, \qquad U = \frac{1}{2}QV.$$

Example 5: capacitor energy

A capacitor has capacitance

$$C = 100 \cdot 10^{-6}\,\mathrm{F}$$

and voltage

$$V = 5\,\mathrm{V}.$$

Find the stored energy.

$$U = \frac{1}{2}CV^2 = \frac{1}{2}(100 \cdot 10^{-6})(5^2).$$

$$U = 50 \cdot 10^{-6} \cdot 25 = 1.25 \cdot 10^{-3}\,\mathrm{J}.$$

So the energy is 1.25 mJ.

5.3 Why capacitors matter in computing

Capacitors appear everywhere in computing hardware:

- smoothing power supply noise,
- storing bits in dynamic RAM,
- filtering signals,
- timing circuits,
- touch sensors,
- analog-to-digital converter input stages.

Computer science viewpoint

A capacitor is a physical memory of voltage. It does not store a variable forever, because real capacitors leak charge. DRAM must refresh stored charge periodically, which is an electromagnetic fact hidden behind memory architecture.

6 Current, resistance, and simple circuits

6.1 Electric current

Electric current is the rate of flow of charge:

$$I = \frac{\Delta Q}{\Delta t}.$$

In calculus notation,

$$I = \frac{\mathrm{d}Q}{\mathrm{d}t}.$$

The unit is the ampere:

$$1\,\mathrm{A} = 1\frac{\mathrm{C}}{\mathrm{s}}.$$

Example 6: charge transported by current

A current of 2 A flows for 3 s. How much charge passes through the wire?

$$Q = It = (2)(3) = 6\,\mathrm{C}.$$

6.2 Ohm's law

For many conductors, voltage and current are approximately proportional:

$$V = IR.$$

Here R is resistance, measured in ohms Ω.

Example 7: Ohm's law

A resistor has $R = 100\Omega$ and voltage $V = 5\,\mathrm{V}$. The current is

$$I = \frac{V}{R} = \frac{5}{100} = 0.05\,\mathrm{A}.$$

That is 50 mA.

Common mistake

Do not confuse current and voltage. Voltage is like an energy difference per charge. Current is flow rate of charge. A circuit can have voltage without much current if resistance is very large.

6.3 Series and parallel resistors

For resistors in series:

$$R_{\mathrm{eq}} = R_1 + R_2 + \cdots + R_n.$$

For resistors in parallel:

$$\frac{1}{R_{\mathrm{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}.$$

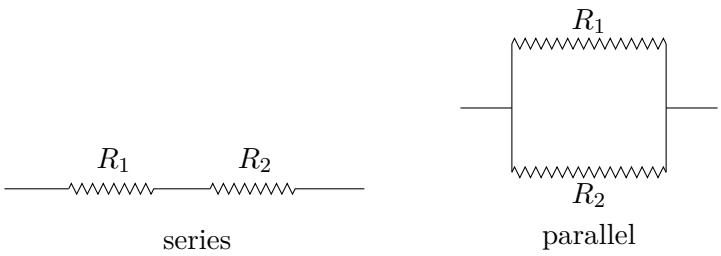


Figure 7: Series and parallel resistor arrangements.

6.4 Kirchhoff's rules

Kirchhoff's current rule:

$$\sum I_{\text{in}} = \sum I_{\text{out}}.$$

This is conservation of charge.

Kirchhoff's voltage rule:

$$\sum \Delta V = 0$$

around a closed loop. This is conservation of energy.

Computer science viewpoint

Circuit solving often becomes linear algebra. Unknown currents or node voltages satisfy a system of equations. A circuit simulator builds a matrix and solves it numerically.

7 RC circuits: the first dynamic circuit

7.1 Charging a capacitor

Consider a resistor R, capacitor C, and battery voltage V_0. When the switch is closed, the capacitor charges gradually.

The voltage across the capacitor is

$$V_C(t) = V_0 \left(1 - e^{-t/(RC)}\right).$$

The product

$$\tau = RC$$

is the time constant.

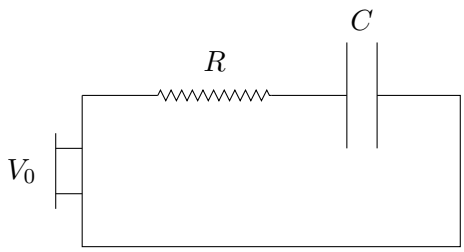


Figure 8: A simple RC charging circuit.

7.2 What the time constant means

At $t = \tau = RC$,

$$V_C(\tau) = V_0(1 - e^{-1}) \approx 0.632V_0.$$

So after one time constant, the capacitor reaches about 63% of its final voltage.

After about 5τ, it is very close to fully charged.

Example 8: RC time constant

Let

$$R = 10\,\mathrm{k\Omega} = 10,000\Omega, \qquad C = 100\,\mu\mathrm{F} = 100 \cdot 10^{-6}\,\mathrm{F}.$$

Then

$$\tau = RC = (10,000)(100 \cdot 10^{-6}) = 1\,\mathrm{s}.$$

So the voltage changes significantly over seconds, not nanoseconds.

7.3 Numerical simulation of an RC circuit

The capacitor equation can be written as

$$\frac{\mathrm{d}V_C}{\mathrm{d}t} = \frac{V_0 - V_C}{RC}.$$

A simple Euler simulation is:

```
V = 0
for step in range(N):
    dVdt = (V0 - V) / (R*C)
    V = V + dt*dVdt
```

Computer science viewpoint

This is the same pattern as many simulations: compute derivative, multiply by small time step, update state. Electromagnetism becomes computational when continuous equations are discretized.

8 Magnetic fields

8.1 What is magnetism?

Magnetic effects are produced by moving electric charges. A steady current in a wire creates a magnetic field around the wire.

The magnetic field is denoted by $\vec{B}$. Its unit is the tesla, T.

Key idea

Electric charges create electric fields. Moving electric charges create magnetic fields.

8.2 Magnetic force on a moving charge

A charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ experiences force

$$\vec{F} = q\vec{v} \times \vec{B}.$$

The magnitude is

$$F = |q|vB\sin\theta,$$

where θ is the angle between $\vec{v}$ and $\vec{B}$.

Important consequences:

- If the particle is not moving, $v = 0$, so magnetic force is zero.
- If the particle moves parallel to $\vec{B}$, then $\theta = 0$, so magnetic force is zero.
- Magnetic force is perpendicular to velocity, so it changes direction of motion rather than directly changing speed.

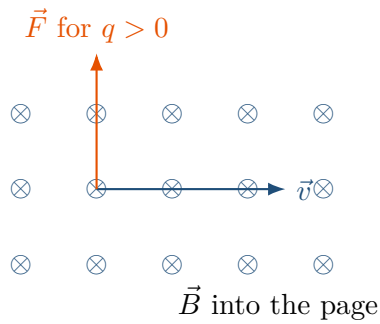


Figure 9: Magnetic force is perpendicular to velocity and magnetic field.

8.3 Right-hand rule

For a positive charge:

1. Point your fingers in the direction of $\vec{v}$.
2. Curl them toward $\vec{B}$.
3. Your thumb gives the direction of $\vec{v} \times \vec{B}$, hence $\vec{F}$.

For a negative charge, the force is in the opposite direction.

Common mistake

The magnetic force direction is one of the most common sources of mistakes. Always check the sign of the charge. Electrons move opposite to conventional current direction.

8.4 Force on a current-carrying wire

For a straight wire of length L carrying current I in magnetic field B, the force magnitude is

$$F = ILB \sin\theta.$$

This is the physical basis of electric motors and speakers.

Example 9: magnetic force on a wire

A wire carries current $I = 2\,\mathrm{A}$, has length $L = 0.5\,\mathrm{m}$, and is perpendicular to a magnetic field $B = 0.1\,\mathrm{T}$. Then

$$F = ILB = (2)(0.5)(0.1) = 0.1\,\mathrm{N}.$$

9 Magnetic fields made by currents

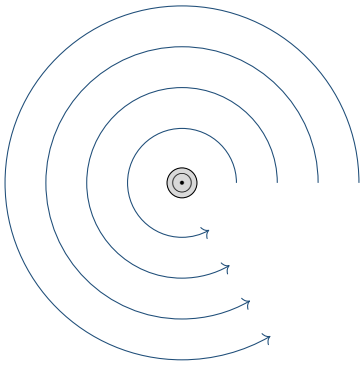
9.1 Field around a long straight wire

A long straight wire carrying current I creates circular magnetic field lines around it. At distance r, the field magnitude is

$$B = \frac{\mu_0 I}{2\pi r},$$

where

$$\mu_0 = 4\pi \cdot 10^{-7}\,\mathrm{N/A^2}.$$



current out of page, magnetic field circles counterclockwise

Figure 10: Magnetic field around a straight current-carrying wire.

9.2 Why this matters for computers

Every current in a computer creates a magnetic field. Usually this field is small, but in fast circuits and dense boards it can cause real problems:

- electromagnetic interference,
- signal crosstalk,
- noise in analog measurements,
- unwanted coupling between traces,
- radiation from high-speed digital lines.

Computer science viewpoint

A digital signal is not pure logic. It is a physical voltage/current waveform propagating along conductors. At high frequencies, wires behave less like ideal connections and more like electromagnetic structures.

9.3 Magnetic dipoles and loops

A current loop acts like a small magnet. Many magnetic materials can be understood as collections of microscopic magnetic moments.

For a loop of area A carrying current I, the magnetic dipole moment magnitude is

$$\mu = IA.$$

10 Electromagnetic induction

10.1 Magnetic flux

Magnetic flux measures how much magnetic field passes through a surface. For uniform field perpendicular to a flat area,

$$\Phi_B = BA.$$

More generally,

$$\Phi_B = \int \vec{B} \cdot \mathrm{d}\vec{A}.$$

For beginners, read this as:

Key idea

Flux = field strength times effective area.

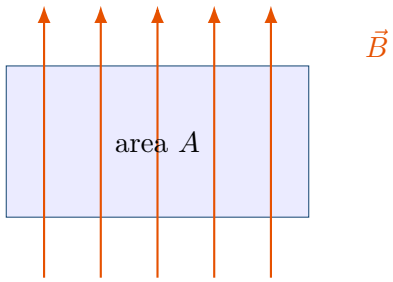


Figure 11: Magnetic flux through a surface.

10.2 Faraday's law

A changing magnetic flux induces voltage:

$$\mathcal{E} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}.$$

Here $\mathcal{E}$ is induced electromotive force, measured in volts.

The minus sign is Lenz's law.

Key idea

The induced effect opposes the change that caused it.

10.3 Simple induction example

Example 10: induced voltage

A coil has $N = 100$ turns. The magnetic flux through one turn changes from 0.020 Wb to 0.005 Wb in 0.10 s. Estimate the magnitude of induced voltage.
For N turns,

$$|\mathcal{E}| = N \left| \frac{\Delta \Phi_B}{\Delta t} \right|.$$

The flux change is

$$\Delta \Phi_B = 0.005 - 0.020 = -0.015\,\text{Wb}.$$

So

$$|\mathcal{E}| = 100 \frac{0.015}{0.10} = 15\,\text{V}.$$

10.4 Inductors

An inductor stores energy in a magnetic field. Its voltage-current relation is

$$V_L = L \frac{\mathrm{d}I}{\mathrm{d}t}.$$

The stored magnetic energy is

$$U = \frac{1}{2} L I^2.$$

Common mistake

A capacitor resists sudden voltage changes. An inductor resists sudden current changes. This sentence helps avoid many circuit mistakes.

11 Maxwell's equations: the complete classical picture

Maxwell's equations are the central laws of classical electromagnetism. They connect electric fields, magnetic fields, charges, and currents.

11.1 Integral form

Law	Equation	Meaning
Gauss's law for electricity	$\oint \vec{E} \cdot \mathrm{d}\vec{A} = \frac{Q_{\text{inside}}}{\varepsilon_0}$	charge creates electric field
Gauss's law for magnetism	$\oint \vec{B} \cdot \mathrm{d}\vec{A} = 0$	no magnetic monopoles in classical EM
Faraday's law	$\oint \vec{E} \cdot \mathrm{d}\vec{l} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}$	changing magnetic field creates electric field
Ampere-Maxwell law	$\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 I_{\text{inside}} + \mu_0 \varepsilon_0 \frac{\mathrm{d}\Phi_E}{\mathrm{d}t}$	current and changing electric field create magnetic field

11.2 What each equation says in plain language

1. Gauss's law for electricity

Electric field lines begin on positive charge and end on negative charge. A closed surface with net charge inside has net electric flux through it.

2. Gauss's law for magnetism

Magnetic field lines do not begin or end. They form closed loops. In classical electromagnetism there are no isolated magnetic charges.

3. Faraday's law

A changing magnetic field creates a circulating electric field. This is the basis of generators, transformers, and many sensors.

4. Ampere-Maxwell law

A current creates magnetic field. Also, a changing electric field creates magnetic field. Maxwell's additional term made electromagnetic waves possible in the theory.

Computer science viewpoint

Maxwell's equations are local update rules for fields. In numerical electromagnetism, one computes how $\vec{E}$ and $\vec{B}$ evolve across a grid. This is very close in spirit to cellular automata or finite-difference simulations, but with physical constraints.

11.3 Differential form: for later courses

Using divergence and curl, Maxwell's equations can be written as

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0},$$

$$\nabla \cdot \vec{B} = 0,$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t},$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}.$$

For absolute beginners, do not panic. These symbols mean:

- $\nabla\cdot$ measures source-like behavior;
- $\nabla\times$ measures circulation or rotation-like behavior;
- ρ is charge density;
- $\vec{J}$ is current density.

12 Electromagnetic waves

12.1 The big idea

Maxwell's equations imply that changing electric and magnetic fields can sustain each other and travel through space as a wave.

Key idea

Light is an electromagnetic wave.

In vacuum, electromagnetic waves travel at speed

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3.00 \cdot 10^8 \,\mathrm{m/s}.$$

12.2 Wave picture

In a plane electromagnetic wave:

- the electric field $\vec{E}$ oscillates,
- the magnetic field $\vec{B}$ oscillates,
- $\vec{E}$ and $\vec{B}$ are perpendicular,
- both are perpendicular to the direction of propagation.

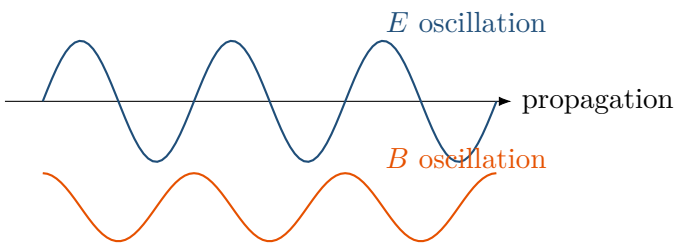


Figure 12: A schematic view of oscillating fields in an electromagnetic wave.

12.3 Frequency, wavelength, and speed

For any wave,

$$v = f\lambda.$$

For electromagnetic waves in vacuum,

$$c = f\lambda.$$

Example 11: wavelength of a Wi-Fi signal

A signal has frequency $f = 2.4\,\mathrm{GHz} = 2.4 \cdot 10^9\,\mathrm{Hz}$. Estimate its wavelength in air.

$$\lambda = \frac{c}{f} = \frac{3.0 \cdot 10^8}{2.4 \cdot 10^9} = 0.125\,\mathrm{m}.$$

So the wavelength is about 12.5 cm.

12.4 The electromagnetic spectrum

Electromagnetic waves include:

- radio waves,
- microwaves,
- infrared,
- visible light,
- ultraviolet,
- X-rays,
- gamma rays.

They differ mainly by frequency and wavelength.

Computer science viewpoint

Wireless communication, optical fibers, RFID, Bluetooth, Wi-Fi, GPS, microwave links, camera sensors, LiDAR, and displays are all electromagnetic technologies. Software often hides this, but hardware never escapes it.

13 Signals, transmission lines, and why wires are not ideal

13.1 The beginner model of a wire

In basic circuits, a wire is treated as a perfect connection. It has no resistance, no inductance, no capacitance, and signals move instantly.

This is a useful fiction for slow circuits.

13.2 The real model of a wire

A real wire has:

- resistance,
- capacitance to nearby conductors,
- inductance from magnetic fields around current,
- finite signal propagation speed.

At high frequencies or fast edge transitions, wires become transmission lines.

Key idea

A digital signal is an electromagnetic wave guided by conductors.

13.3 Impedance intuition

Resistance describes how a circuit opposes steady current. Impedance generalizes this idea to time-varying signals.

Capacitors and inductors have frequency-dependent behavior:

$$X_C = \frac{1}{2\pi f C},$$

$$X_L = 2\pi f L.$$

A capacitor is easier for high frequencies to pass. An inductor is harder for high frequencies to pass.

Example 12: capacitor reactance

Let $C = 1\,\mu\text{F}$.
At $f = 100\,\text{Hz}$,

$$X_C = \frac{1}{2\pi(100)(10^{-6})} \approx 1592\Omega.$$

At $f = 100\,\text{kHz}$,

$$X_C = \frac{1}{2\pi(100000)(10^{-6})} \approx 1.59\Omega.$$

The same capacitor behaves very differently at different frequencies.

13.4 Why this matters for computer systems

In high-speed digital design:

- clock edges contain high-frequency components;
- PCB traces can reflect signals;
- impedance mismatch can create ringing;
- ground bounce can corrupt logic levels;
- electromagnetic interference can create bugs that look random.

14 Numerical electromagnetism for programmers

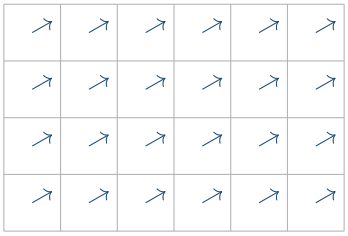
14.1 Discretizing space

Continuous space has infinitely many points. A computer cannot store infinitely many field values. So we replace space by a grid.

For example, in two dimensions:

$$E_x[i,j], \quad E_y[i,j], \quad B_z[i,j].$$

The indices i, j identify grid cells.



field values stored on a grid

Figure 13: A vector field discretized on a rectangular grid.

14.2 Finite differences

A derivative can be approximated by a difference quotient:

$$\frac{\partial f}{\partial x} \approx \frac{f[i+1] - f[i]}{\Delta x}.$$

This is how continuous equations become executable code.

```
for i in range(N-1):
    derivative[i] = (f[i+1] - f[i]) / dx
```

14.3 A very simplified field update idea

Maxwell's equations tell us that changing B updates E, and changing E updates B. A symbolic simulation loop looks like:

```
for each time step:
    update_E_from_spatial_changes_of_B()
    apply_sources_and_boundaries()
    update_B_from_spatial_changes_of_E()
```

This is not a complete production solver, but it shows the algorithmic structure.

Common mistake

Numerical electromagnetism is sensitive to grid size, time step, and boundary conditions. A simulation can produce beautiful nonsense if the discretization is unstable or the physical assumptions are wrong.

14.4 Stability intuition

If information physically travels at speed c, a numerical method must not take time steps that jump too far across the grid. A rough idea is

$$\Delta t \lesssim \frac{\Delta x}{c}.$$

This is related to the Courant condition in wave simulations.

15 Complete worked examples

15.1 Worked example A: from voltage to acceleration

Electron in a uniform electric field

An electron is placed in a uniform electric field of magnitude

$$E = 1000\,\mathrm{N/C}.$$

Find the magnitude of the force and acceleration.
Electron charge magnitude:

$$|q| = 1.60 \cdot 10^{-19}\,\mathrm{C}.$$

Electron mass:

$$m = 9.11 \cdot 10^{-31}\,\mathrm{kg}.$$

Force magnitude:

$$F = |q|E = (1.60 \cdot 10^{-19})(1000) = 1.60 \cdot 10^{-16}\,\mathrm{N}.$$

Acceleration magnitude from Newton's second law:

$$a = \frac{F}{m} = \frac{1.60 \cdot 10^{-16}}{9.11 \cdot 10^{-31}} \approx 1.76 \cdot 10^{14}\,\mathrm{m/s^2}.$$

This is enormous compared with everyday accelerations. Small particles can accelerate very rapidly under electric fields.
Because the electron is negative, its acceleration is opposite to the electric field direction.

15.2 Worked example B: solving a simple node voltage

Voltage divider

Two resistors are connected in series across $V_{in} = 12\,\mathrm{V}$:

$$R_1 = 2\,\mathrm{k\Omega}, \qquad R_2 = 4\,\mathrm{k\Omega}.$$

The output voltage is measured across R_2. Find V_{out}.
Total resistance:

$$R_{tot} = R_1 + R_2 = 6\,\mathrm{k\Omega}.$$

Current:

$$I = \frac{V_{in}}{R_{tot}} = \frac{12}{6000} = 0.002\,\mathrm{A}.$$

Voltage across R_2:

$$V_{out} = IR_2 = (0.002)(4000) = 8\,\mathrm{V}.$$

Shortcut formula:

$$V_{out} = V_{in}\frac{R_2}{R_1 + R_2} = 12\frac{4}{6} = 8\,\mathrm{V}.$$

15.3 Worked example C: magnetic bending

Charged particle in a magnetic field

A positive particle with charge q, mass m, and speed v enters a uniform magnetic field B perpendicular to its velocity. Show that it moves in a circle and find the radius.
The magnetic force magnitude is

$$F = qvB.$$

Because the force is perpendicular to velocity, it acts as centripetal force:

$$F = \frac{mv^2}{r}.$$

Equate the two:

$$qvB = \frac{mv^2}{r}.$$

Cancel one factor of v:

$$qB = \frac{mv}{r}.$$

Solve for r:

$$r = \frac{mv}{qB}.$$

So faster particles curve less, stronger magnetic fields curve more, and heavier particles curve less.

16 Typical mistakes and how to avoid them

Common mistake

Mistake 1: treating vectors like ordinary numbers.
Force, field, velocity, and magnetic field have directions. You cannot simply add magnitudes unless all vectors point along the same line.

Common mistake

Mistake 2: forgetting the sign of charge.
Electric field direction is defined by the force on a positive test charge. An electron feels force opposite to $\vec{E}$.

Common mistake

Mistake 3: confusing electric field and electric potential.
Electric field is a vector. Potential is a scalar. Voltage difference is not the same object as field, although they are related.

Common mistake

Mistake 4: assuming wires are ideal at all frequencies.
At high speed, traces and cables have capacitance, inductance, delay, reflection, and radiation. Digital logic depends on analog electromagnetic behavior.

Common mistake

Mistake 5: trusting simulations without checking units and scales.
A simulation with wrong units may still run. Running code is not the same as correct physics.

17 Formula sheet

Electrostatics

$$F = k\frac{|q_1 q_2|}{r^2}, \qquad k = \frac{1}{4\pi\varepsilon_0}.$$

$$\vec{E} = \frac{\vec{F}}{q}, \qquad \vec{F} = q\vec{E}.$$

$$E = k\frac{|Q|}{r^2}.$$

$$V = \frac{U}{q}, \qquad \Delta U = q\Delta V.$$

$$E \approx \frac{|\Delta V|}{d}.$$

Circuits

$$I = \frac{\Delta Q}{\Delta t}, \qquad V = IR.$$

$$R_{series} = R_1 + R_2 + \cdots .$$

$$\frac{1}{R_{parallel}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots .$$

$$Q = CV, \qquad U_C = \frac{1}{2}CV^2.$$

$$V_C(t) = V_0(1 - e^{-t/(RC)}), \qquad \tau = RC.$$

Magnetism and induction

$$\vec{F} = q\vec{v} \times \vec{B}, \qquad F = |q|vB\sin\theta.$$

$$F = ILB\sin\theta.$$

$$B = \frac{\mu_0 I}{2\pi r}.$$

$$\Phi_B = \int \vec{B} \cdot \mathrm{d}\vec{A}, \qquad \mathcal{E} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t}.$$

$$V_L = L\frac{\mathrm{d}I}{\mathrm{d}t}, \qquad U_L = \frac{1}{2}LI^2.$$

Waves

$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}, \qquad v = f\lambda, \qquad c = f\lambda \quad \text{in vacuum.}$$

18 Practice problems

A. Charge and electric field

1. Two charges $q_1 = 1 \cdot 10^{-6}\,\mathrm{C}$ and $q_2 = 4 \cdot 10^{-6}\,\mathrm{C}$ are separated by $0.30\,\mathrm{m}$. Find the force magnitude.
2. A charge $q = 5 \cdot 10^{-6}\,\mathrm{C}$ is placed in a field $\vec{E} = (100, 200)\,\mathrm{N/C}$. Find $\vec{F}$.
3. A voltage difference of $9\,\mathrm{V}$ occurs over $3\,\mathrm{mm}$. Estimate the electric field magnitude.

B. Circuits

1. A 220Ω resistor is connected to $5\,\mathrm{V}$. Find the current.
2. Two resistors 100Ω and 300Ω are in series. Find the equivalent resistance.
3. The same resistors are in parallel. Find the equivalent resistance.
4. A capacitor $C = 10\,\mu\mathrm{F}$ is charged to $12\,\mathrm{V}$. Find the stored energy.
5. For $R = 1\,\mathrm{k}\Omega$ and $C = 100\,\mu\mathrm{F}$, find the RC time constant.

C. Magnetism and induction

1. A wire of length $0.2\,\mathrm{m}$ carries $3\,\mathrm{A}$ perpendicular to a $0.4\,\mathrm{T}$ field. Find the force.
2. A charge $q = 2 \cdot 10^{-6}\,\mathrm{C}$ moves at $1000\,\mathrm{m/s}$ perpendicular to $B = 0.5\,\mathrm{T}$. Find the magnetic force magnitude.
3. A coil of 50 turns has flux change $0.01\,\mathrm{Wb}$ over $0.2\,\mathrm{s}$. Estimate the induced voltage magnitude.

D. Waves

1. Find the wavelength of a $5\,\mathrm{GHz}$ electromagnetic wave.
2. A radio wave has wavelength $3\,\mathrm{m}$. Find its frequency.

19 Selected solutions

A1

$$F = k\frac{|q_1 q_2|}{r^2} = (8.99 \cdot 10^9)\frac{(1 \cdot 10^{-6})(4 \cdot 10^{-6})}{(0.30)^2}.$$
$$F \approx 0.40\,\mathrm{N}.$$

A2

$$\vec{F} = q\vec{E} = (5 \cdot 10^{-6})(100, 200) = (5 \cdot 10^{-4}, 10^{-3})\,\mathrm{N}.$$

A3

$$E \approx \frac{9}{0.003} = 3000\,\mathrm{V/m}.$$

B1

$$I = \frac{V}{R} = \frac{5}{220} \approx 0.0227\,\mathrm{A} = 22.7\,\mathrm{mA}.$$

B2

$$R_{eq} = 100 + 300 = 400\Omega.$$

B3

$$\frac{1}{R_{eq}} = \frac{1}{100} + \frac{1}{300} = \frac{4}{300}.$$
$$R_{eq} = 75\Omega.$$

B4

$$U = \frac{1}{2}CV^2 = \frac{1}{2}(10 \cdot 10^{-6})(12^2) = 7.2 \cdot 10^{-4}\,\mathrm{J}.$$

B5

$$\tau = RC = (1000)(100 \cdot 10^{-6}) = 0.1\,\mathrm{s}.$$

C1

$$F = ILB = (3)(0.2)(0.4) = 0.24\,\mathrm{N}.$$

C2

$$F = |q|vB = (2 \cdot 10^{-6})(1000)(0.5) = 10^{-3}\,\mathrm{N}.$$

C3

$$|\mathcal{E}| = N\left|\frac{\Delta\Phi}{\Delta t}\right| = 50\frac{0.01}{0.2} = 2.5\,\mathrm{V}.$$

D1

$$\lambda = \frac{c}{f} = \frac{3.0\cdot 10^8}{5\cdot 10^9} = 0.06\,\mathrm{m} = 6\,\mathrm{cm}.$$

D2

$$f = \frac{c}{\lambda} = \frac{3.0\cdot 10^8}{3} = 1.0\cdot 10^8\,\mathrm{Hz} = 100\,\mathrm{MHz}.$$

20 Final conceptual map

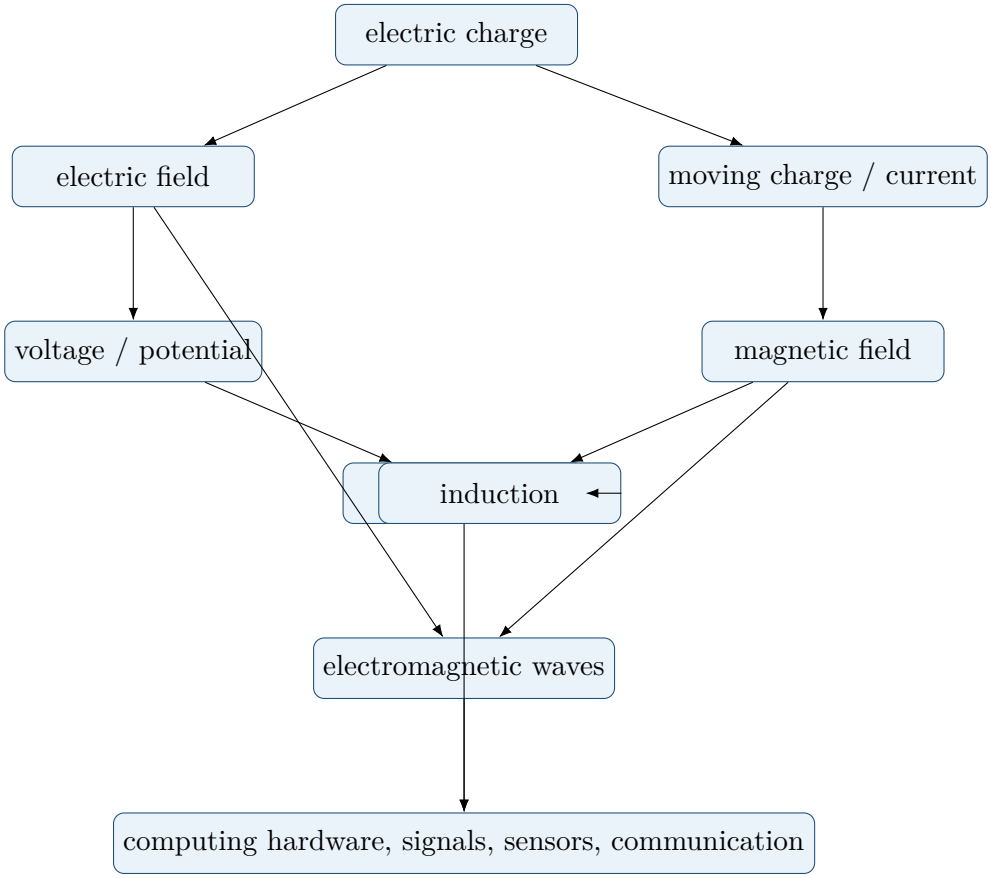


Figure 14: A map of the main ideas in introductory electromagnetism.

Key idea

Electromagnetism is not a list of unrelated formulas. It is a connected theory: charge creates fields, currents create magnetic effects, changing fields create other fields, and the combined dynamics produce waves and real electronic behavior.